Javier Senabre Urrea



Computer Engineer | AI, Data and Software Development

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Professional Summary

Computer engineer with a Master's degree and a Big Data and Cloud Computing track. I turn data into decisions with machine learning and build the software around it: from a clustering thesis on EHF data (76–79% accuracy) and a bachelor's thesis that classifies birds by their song, to a chatbot with human handover in Chatwoot and virtual reality games. Looking for a data or development team where I can keep growing.

Projects

Master's Thesis: profiling high-performance handball players (EHF EURO 2024) | 2025 – 2026

- Clustering (K-Means, Ward, DBSCAN) on 19 per-player variables, with 80% agreement with MVP nominations.
- Random Forest, XGBoost, SVM and MLP tuned with GridSearchCV inside Pipelines: 76% (men) and 79% (women) accuracy. Interpretability with SHAP.
- Sprint-based repository (Scrum); 107-page thesis in LaTeX. Stack: Python, pandas, scikit-learn, XGBoost, SHAP, Jupyter.

Bachelor's Thesis: bird song classifier | 2023 – 2024

- Classifier for 50 bird species from recordings of their song, using an Xception convolutional network on spectrograms: 72% accuracy. Dataset of more than 2,000 recordings downloaded from Xeno-canto, respecting its licence and terms of use.
- Audio processing and mel spectrogram generation as model input, using windowing to remove silences and data augmentation.
- Flask web app to classify an audio file from the browser.
- Stack: Python, TensorFlow/Keras, librosa, Flask.

Artificial bird song generation with diffusion | Ongoing personal project | 2026

- Artificial generation of bird songs from mel spectrograms with diffusion models.
- Data pipeline: downloading and preparing Xeno-canto recordings and converting them to mel spectrograms.
- A conditional DDPM trained from scratch, with audio reconstruction through a vocoder.
- Generates audio very close to real recordings for some species, such as the common chiffchaff.
- Stack: Python, PyTorch, Stable Diffusion.

Work Experience

Computer Engineer | Panadero SL | 02/2025 – 05/2025 (3 months)

- Built a working chatbot that answers customer questions about the company and its services: requirements analysis and definition, process automation with n8n, REST API integration and testing.
- Integrated Chatwoot, a chat application that hands over to a person the queries the chatbot cannot resolve. Stack: n8n, REST API, Chatwoot.

Virtual Reality Game Developer | Cerebroom VR Games, Albacete | 02/2024 – 05/2024 (3 months)

- Developed virtual reality games in C# within a Scrum team.

Education

Master's Degree in Computer Engineering | University of Castilla-La Mancha (UCLM), Albacete | 2024 – 2026

- **Big Data and Cloud Computing track**

Bachelor's Degree in Computer Engineering | UCLM, Albacete | 2020 – 2024

Technical Skills

Languages: Python, C, Java, C#, JavaScript, SQL, HTML, CSS

Machine learning and data: clustering, supervised classification, deep learning (CNN), SHAP, pandas, NumPy, scikit-learn, TensorFlow/Keras, PyTorch, Power BI, Grafana

Development, Big Data and cloud: REST API, microservices, Flask, Spark, MongoDB (NoSQL), AWS, Docker, n8n, Git/GitHub, Linux, Arduino and ESP32

Methodologies and soft skills: Scrum, requirements engineering, testing, communication, project organization, teamwork

Languages and Additional Information

Languages: Spanish (native), English (B1). Additional: quantum computing course (UCLM), B driving licence. Interests: bird watching and recording, the origin of my audio projects.